

Satellite-Based Monitoring of Slow Ground Deformation in a Zimbabwean Slope-Terrain Study Area: An InSAR and Machine Learning Approach

A Project Proposal Submitted for Consideration

Group Members

Midlands State University
Department of Mathematics and Computational Sciences
BSc Applied Mathematics and Computational Science

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Abstract

This proposal outlines a project to monitor slow ground deformation in a slope-terrain area of Zimbabwe using Sentinel-1 satellite radar imagery, processed through the Small Baseline Subset (SBAS) InSAR technique, combined with a statistical anomaly detection layer to flag deformation behaviour of potential concern. The approach is well established internationally and has recently been demonstrated in comparable East African terrain (Kenya), but has not, to the group's knowledge, been applied to a Zimbabwean study area. The project aims to produce a validated displacement time series, a working anomaly flagging method, and an assessment of the technique's applicability to local terrain and land cover conditions.

1 Introduction and Background

Regions of Zimbabwe with steep or sloped terrain host critical infrastructure, including roads, residential structures, and rural settlements, that are subject to slow, ongoing ground deformation. This deformation arises from processes such as soil creep, rainfall-induced slope instability, and localized subsidence, and typically proceeds at rates of a few millimetres to centimetres per year. Because this movement is gradual and largely invisible to routine observation, monitoring in practice tends to be reactive: intervention occurs only after visible damage, such as road cracking or slope failure, has already occurred.

Interferometric Synthetic Aperture Radar (InSAR), applied to freely available Sentinel-1 satellite imagery, is an established remote sensing technique capable of detecting surface displacement at millimetre-scale precision over wide areas and extended time periods. While InSAR-based deformation monitoring has been applied extensively in regions such as East Asia, the Himalayas, and parts of North Africa, its application to Zimbabwean terrain remains largely absent from the published literature. This proposal outlines a project to apply InSAR time-series analysis, combined with a statistical and machine-learning-based anomaly detection layer, to a defined small-scale study area within Zimbabwe.

2 Problem Statement

There is currently no continuous, quantitative monitoring system for slow-onset ground deformation in Zimbabwean slope-terrain regions bearing infrastructure. Existing risk assessment relies on visual inspection and post-incident response, which by nature cannot anticipate deformation-driven failures before they occur. This project addresses that gap by developing a reproducible pipeline that (i) quantifies surface displacement over time using satellite radar data, and (ii) applies anomaly detection to flag deformation behaviour that deviates from an area's established pattern, providing an early indicator of elevated risk.

3 Aim and Objectives

Aim: To develop and validate a satellite-based ground deformation monitoring and anomaly detection pipeline for a defined study area in Zimbabwe.

Objectives:

- i. To acquire and process a Sentinel-1 SAR image stack for the selected study area using an established InSAR time-series method (Small Baseline Subset, SBAS).
- ii. To generate a displacement time series for coherent surface points within the study area.

- iii. To develop and apply a statistical and/or machine-learning-based anomaly detection method to identify deformation behaviour inconsistent with established baseline trends.
- iv. To validate flagged anomalies against optical imagery and any available secondary records (e.g. reported road damage or slope incidents).
- v. To assess the applicability and limitations of the InSAR approach specifically for Zimbabwean terrain and land cover conditions.

4 Significance of the Study

This project is significant for three reasons. First, it addresses a genuine infrastructure safety concern relevant to road and slope engineering in Zimbabwe, an area with limited existing monitoring infrastructure. Second, it contributes an applied case study of InSAR deformation monitoring in a geographic and geological setting, the Zimbabwean Precambrian craton and associated terrain, that is underrepresented in current literature, most published InSAR case studies concentrating on Asian, European, or geologically active regions. Third, the project offers a natural extension pathway: the same pipeline can, with modification, be applied to monitoring river-corridor change or fault-associated features, giving the work value beyond its immediate scope.

5 Literature Context

InSAR time-series analysis has become a standard geodetic tool for measuring slow surface displacement, with documented millimetre-scale accuracy across applications including land subsidence, landslide monitoring, and mining-related ground movement [berardino2002](#). The Small Baseline Subset (SBAS) method in particular is well suited to natural, non-urban terrain, where classical persistent scatterer methods are less effective due to lower point density [lanari2007](#).

Of direct relevance to this proposal, SBAS-InSAR has recently been applied to landslide-prone terrain in Murang'a and Nyeri Counties, Kenya [kenya2025](#), an East African setting with broadly comparable land cover, rainfall-driven slope instability, and limited existing monitoring infrastructure. This study demonstrates that the technique is both technically viable and practically informative in an African context similar to the proposed study area, and serves as the closest regional precedent available. Complementary work combining InSAR-derived deformation fields with a secondary statistical or machine-learning analysis layer has also been demonstrated in Quetta, Pakistan [quetta2024](#) and Tianjin, China [tianjin2025](#), both showing that layering anomaly detection on top of raw deformation output improves the practical usefulness of the results for hazard identification. No published study, to the group's knowledge, has yet applied this combined approach to a Zimbabwean study area; this project addresses that specific gap.

6 Methodology

6.1 Data Acquisition

Sentinel-1 Single Look Complex (SLC) imagery will be acquired for the selected area of interest, with a minimum of fifteen to twenty acquisitions spanning at least one year, in order to ade-

quately sample seasonal atmospheric variation. A Copernicus GLO-30 Digital Elevation Model will be used for topographic phase correction.

6.2 InSAR Time-Series Processing

Processing will follow the SBAS methodology, implemented using the open-source MintPy package. For a pair of SAR acquisitions at times t_A and t_B , the differential interferometric phase is modelled as

$$\delta\phi = \frac{4\pi}{\lambda} \Delta d_{LOS} + \delta\phi_{\text{topo}} + \delta\phi_{\text{atm}} + \delta\phi_{\text{noise}},$$

where Δd_{LOS} is the line-of-sight surface displacement between acquisitions, λ is the Sentinel-1 wavelength (5.55 cm), and $\delta\phi_{\text{topo}}$ and $\delta\phi_{\text{atm}}$ represent residual topographic and atmospheric phase contributions, corrected using the DEM and standard atmospheric filtering respectively.

Because measured phase is recorded modulo 2π , a phase unwrapping step (minimum cost flow) is applied to recover continuous phase values. Given N SAR acquisitions and M interferometric pairs formed under small spatial and temporal baseline constraints, the unknown mean velocities $\mathbf{v}$ between consecutive acquisitions are recovered via the linear system

$$\mathbf{A}\mathbf{v} = \frac{\delta\phi}{4\pi/\lambda},$$

solved in the least-squares sense as

$$\hat{\mathbf{v}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \left(\frac{\delta\phi}{4\pi/\lambda} \right),$$

with cumulative displacement obtained by temporal integration of $\hat{\mathbf{v}}$.

6.3 Anomaly Detection

For each coherent pixel, a baseline deformation trend $\hat{d}_i(t)$ is estimated from the displacement time series $d_i(t)$. A point is flagged as anomalous where

$$|d_i(t) - \hat{d}_i(t)| > k\sigma_i,$$

with σ_i the pixel's characteristic deformation variability and k a threshold (typically 2–3), determined empirically. Should time permit, this statistical baseline will be extended with a machine-learning approach (e.g. an autoencoder trained on normal deformation patterns, with reconstruction error used as an anomaly score), allowing comparison between the two methods.

6.4 Validation

Flagged anomalies will be cross-referenced against Sentinel-2 optical imagery and any available reports of road or slope damage within the study area, to assess the practical reliability of the detection method.

7 Timeline

Phase	Duration
Study area selection and scoping	Week 1
Data acquisition (SAR imagery, DEM)	Weeks 2–3
InSAR processing (SBAS pipeline)	Weeks 4–6
Anomaly detection development	Weeks 7–9
Validation and report writing	Weeks 10–11

8 Expected Outcomes

The project is expected to produce: a validated displacement time series for the chosen study area; a working anomaly detection method flagging deformation of potential concern; an assessment of InSAR data quality and coherence characteristics specific to Zimbabwean land cover; and a final report documenting methodology, results, and limitations, suitable as a foundation for further work extending the approach to additional sites or applications.

9 Limitations

Coherence loss in areas of dense vegetation is a known limitation of InSAR over natural terrain and is expected to reduce point density in parts of the study area. Atmospheric phase delay remains a source of residual noise despite correction. These limitations will be explicitly assessed and reported rather than treated as a threat to the validity of the overall approach.

References

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Suggested Reading Order for the Group

1. Osmanoglu et al. (2016) — start here. A tutorial-style review of InSAR time-series methods; gives the full landscape before narrowing in.
2. Berardino et al. (2002) — the original SBAS paper. Dense, but this is the source of the exact equations used in this proposal.
3. Lanari et al. (2007) — a more digestible overview of SBAS, written after the method was established.
4. Kenya study (Murang'a and Nyeri Counties) — the closest regional precedent; read this to see what a finished study in comparable terrain looks like.
5. Quetta, Pakistan and Tianjin, China studies — for the anomaly/machine-learning layer specifically.
6. MintPy documentation (GitHub) — once ready to start processing, not before.